

Assignment 1

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MGMT1212: Incident Handling and Response

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February 13, 2026

Cybersecurity Frameworks

HIPAA

What does HIPAA cover?

HIPAA covers who follows the law, the information that it protects, and how information is used and shared. HIPAA applies to organizations that are covered entities and business associates which are healthcare providers, health plans, healthcare clearinghouses, third-party vendors, professional services, data services, and specialized services. HIPAA applies to most organizations that handle and deliver health information but does not apply to all.

HIPAA protects protected health information (PHI) such as medical records and appointment information that can contain contact details and full-face photos. HIPAA requires that the use of PHI is only for allowed purposes and is limited to the least amount of information needed for the purpose. The framework also requires safeguards for PHI and notifications if a breach may occur. Individuals under HIPAA have the right for access of their health information, requests for correction and restrictions for their information, and notices of privacy practices that explains how their information is used and shared.

Why would an organization choose HIPAA?

HIPAA is required by U.S federal law if an organization is a healthcare provider. Hospitals or clinics often demand HIPAA protection in contracts to do business operations. HIPAA also encourages organizations to implement safeguards to increase security and decrease the chances of breaches for PHI as implementing HIPAA can also reduce legal penalties, regulatory fines, and financial damage from breaching.

Organizations may also choose HIPAA to show that health information of patients is handled properly with high transparency which increases confidence and loyalty to their customers. HIPAA also provides guidelines for privacy, security, and documentation that help staff with their operations and lowering confusion or risk. HIPAA compliance generally helps attract patients and customers who want their data protected.

MITRE ATT&CK

What does MITRE ATT&CK cover?

MITRE ATT&CK covers adversary behavior and is a knowledge base of tactics, techniques, and procedures. This framework covers the entire attack chain including reconnaissance, access, persistence, lateral movement, and more. This framework includes attacks against traditional IT and cloud environments, android and IOS devices, and industrial control systems and operational technology.

ATT&CK lists techniques for each tactic such as initial access, persistence, and defense evasion for phishing, application exploits, and abusing valid accounts. ATT&CK helps defenders with detecting or preventing techniques from threat actors by saving entries from threat groups, malwares, and tools while suggesting mitigations and detection ideas from these entries.

Why would an organization choose MITRE ATT&CK?

An organization would choose MITRE ATT&CK because it shares attacker behavior that helps security operations teams build their security programs. ATT&CK removes creating their security around theoretical scenarios or controls driven by compliance frameworks. The framework also helps create mitigations for attack techniques that are more likely and have greater impact to the environment of an organization.

For example, red teams or penetration testers can use ATT&CK to emulate realistic adversary attacks, exercising real attacker playbooks to then measure the effectiveness of controls or blocking and producing repeatable results from each exercise. With this method, ATT&CK provides security and enhanced engineering by describing and sharing attacks and defenses. The collaboration to develop and maintain ATT&CK helps teams understand risks and develop information about threats while continuously improving defenses.

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